Classification aims to predict discrete class labels by assigning an input instance to one of a finite set of predefined categories. The output is categorical rather than numerical. Classification is appropriate when the task involves decision making between distinct classes.
Examples include handwritten digit recognition (predicting digits 0–9), email spam detection (spam or not spam), and medical diagnosis where patients are classified as having or not having a disease.

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Regression, in contrast, aims to predict a continuous numerical value by learning a functional relationship between input features and a real-valued output. Regression is suitable when the target variable is quantitative.
Examples include predicting house prices based on features such as size and location, estimating temperature or rainfall, and forecasting sales or demand over time.

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2019-2020

In summary, classification is used when the output is discrete, while regression is used when the output is continuous, and the choice between the two depends on the nature of the prediction task.

MACHINE LEARNING

Time allowed: 2 hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer All Questions

Only silent, self contained calculators with a Single-Line Display or Dual-Line Display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

a) Classification and regression are two supervised learning techniques used for prediction. Classification aims to predict discrete class labels by assigning each input to one of a finite set of predefined categories, while regression aims to predict continuous numerical values by learning a functional relationship between input variables and a continuous output.

Classification is appropriate when the output variable is categorical. For example, email spam detection classifies emails as spam or non-spam, and medical diagnosis classifies patients as having a disease or not.

Regression is appropriate when the output variable is continuous. For example, house price prediction estimates a numerical price based on features such as size and location, and temperature forecasting predicts a continuous temperature value.

Deep Learning (with respect to Representation Learning)
Deep learning refers to a class of machine learning methods that learn hierarchical representations of data by using multiple layers of nonlinear processing units.
With respect to representation learning, deep learning automatically learns increasingly abstract and high-level features from raw input data, rather than relying on manually designed features.
In a typical deep neural network:
•Lower layers learn simple features (e.g. edges or pixels),
•Middle layers learn more complex patterns (e.g. shapes),
•Higher layers learn task-specific, abstract representations (e.g. object or digit identity).
Thus, deep learning enables the system to learn feature representations and the classifier jointly.
Two Problems When Going Deeper
1️⃣ Vanishing Gradient Problem
The vanishing gradient problem occurs during back-propagation when gradients become very small as they are propagated through many layers.
As a result:
•Early layers receive extremely small gradient updates,
•Learning becomes very slow or may stop completely,
•The network fails to effectively learn deep representations.
This problem is especially severe when using activation functions such as sigmoid or tanh, whose derivatives are small.
2️⃣ Parameter Explosion
Parameter explosion refers to the rapid increase in the number of trainable parameters as the network depth and width increase.
This leads to:
•Higher computational and memory costs,
•Increased risk of over-fitting,
•More difficulty in training due to limited data.
For example, fully connected deep networks can contain millions of parameters, making training inefficient and unstable.

Summary Sentence (Very Useful in Exams)
Deep learning performs representation learning by learning hierarchical features through multiple layers, but increasing depth introduces challenges such as vanishing gradients, which hinder effective training, and parameter explosion, which increases computational cost and over-fitting risk.

Question 1: **[18 marks]**

(a) Discuss the concept of **classification** and **regression** and provide concrete examples when each method is appropriate.

[8 marks]

(b) Referring to the lecture notes, what is deep learning (with respect to representation learning)? Briefly describe the two problems occur when we go deeper, vanishing gradient and parameter explosion. You may answer the first question by drawing a typical deep learning architecture with explanation.

[10 marks]

Question 2: **[20 marks]**

(c) We will use a single ADLINE classifier to classify 2D real number input into two classes (+1 and -1). The training data is listed as follows.

Input	Output
(1,2)	1
(-2,3)	1
(1,-2)	-1
(-2,-1)	-1

Train the ADLINE with online mode using the training data above from the first row to the last row, with the initialization of weights $w=[1\ 0\ 0]^T$ and $\eta=0.5$. For each iteration, calculate the current output of ADLINE (o) and the updated weights (w). Note that you need to show intermediate steps of your calculation. An answer without the intermediate steps will not receive any marks

Online mode (incremental learning) updates the weights after each individual training sample is presented. For each sample, the error is computed and the weights are immediately adjusted using the delta rule. This allows the model to adapt quickly to new data and is suitable for streaming or large datasets, but the learning process may be noisy and less stable.

[16 marks]

(d) What is the difference between online mode and batch mode when training ADLINE?

Batch mode, in contrast, computes the error for all training samples first and updates the weights once per epoch using the sum (or average) of all error gradients. This results in smoother and more stable convergence but requires more computation per update and cannot adapt immediately to individual samples.

[4 marks]

Question 3: **[24 marks]**

In a laparoscopic surgical tool detection task, it is known that the surgical tool components come from one of two classes, C_1 or C_2. Each sample of a surgical tool component Y has two features, $Y = (y_1, y_2)$. An experiment has collected 6 samples and their feature values and classification are shown in the following table. Answer the following questions by showing details of intermediate steps. An answer without the intermediate steps will not receive any marks.

$Y = (y_1, y_2)$		Classification
y_1	y_2	C
3	3	C_2
8	8	C_1
3	5	C_2
5	4	C_2

Idea of Soft Margin SVM
A soft margin Support Vector Machine (SVM) extends the hard-margin SVM by allowing some training samples to violate the margin constraints. Instead of requiring perfect linear separability, soft-margin SVM permits a small number of misclassifications while still aiming to maximise the margin between classes.

Problem It Tries to Tackle
Hard-margin SVM works only when data are perfectly linearly separable.
In real-world applications, data often contain noise, outliers, or overlapping classes, making strict separation impossible. In such cases, a hard-margin SVM fails to find a feasible solution.
How the Problem is Solved
Soft-margin SVM introduces slack variables to measure the degree of constraint violation for each sample. A regularisation parameter C is used to control the trade-off between:
•Maximising the margin, and
•Minimising classification errors.
The optimisation problem balances margin size and misclassification penalty, allowing a robust decision boundary even when the data are not perfectly separable.
One-sentence exam summary
Soft-margin SVM addresses non-linearly separable and noisy data by allowing margin violations through slack variables and balancing margin maximisation and error minimisation using a regularisation parameter.

6	5	C_1
7	6	C_1
6	3.5	=?

(a) Based on the data with known classification, use a KNN classifier with K=3 by the distance metric to classify the unknown sample Y=(6, 3.5) in the last row of the table.

[8 marks]

(b) Suppose the data in the above table are collected in a way that the classification is unknown, i.e., we do not know C in the last column. Cluster the data into two clusters using K-Means algorithm with initial cluster centers Z_1=(3, 3) and Z_2=(5, 4).

Forget Gate
The forget gate controls how much of the previous cell state should be retained or discarded. It takes the previous hidden state and current input as input and outputs a value between 0 and 1 using a sigmoid function. A value close to 1 means the information is kept, while a value close to 0 means the information is forgotten. This mechanism allows the LSTM to remove irrelevant or outdated information from the memory cell.

[16 mark]

Question 4: [38 marks]

Input Gate
The input gate determines how much new information should be written into the cell state. It consists of two components: a sigmoid gate that decides which values should be updated, and a tanh layer that generates candidate new information. The input gate controls whether the new input is important enough to be stored in the memory cell.

(a) Briefly describe the steps to design a learning system according to the lecture notes.

[10 marks]

(b) Briefly describe the idea of soft margin SVM. What is the problem it tries to tackle and how the problem is solved.

One-sentence summary (useful in exams)
The forget gate decides what information to discard from the cell state, while the input gate decides what new information should be added to the memory.

[6 marks]

(c) The formula used for Bayesian learning is as follows. Describe what do the terms (variables) in the formula correspond to in the context of machine learning.

$$P(h|x) = \frac{P(x|h)P(h)}{P(x)}$$

Pre-training and Fine-tuning in Deep Learning
Pre-training is the initial training phase where a deep model is trained on a large dataset (often unsupervised or weakly supervised) to learn general and reusable feature representations. During this phase, the model captures low-level and mid-level patterns that are broadly useful across tasks.
Fine-tuning is the subsequent phase where the pre-trained model is further trained on a task-specific dataset, typically with supervision. The model's parameters are adjusted to specialise the learned representations for the target task.
Why This Helps Model Performance
Pre-training provides a good parameter initialization, which:
reduces the risk of poor local minima,
alleviates the vanishing gradient problem,
speeds up convergence during training.

[10 marks]

h	Hypothesis, class label
x	observed input data, typically represented as a feature vector.
$P(h)$	prior belief (probability of hypothesis before seeing any data)
$P(x)$	∑_h▒〖P(x │ h)P(h)〗 : data evidence (marginal probability of the data)
$P(x\|h)$	likelihood (probability of the data if the hypothesis is true)
$P(h\|x)$	posterior (probability of hypothesis after having seen the data)

Fine-tuning adapts the general features to the specific task, improving accuracy, especially when labelled data are limited, and reducing over-fitting.

One-sentence exam summary
Pre-training learns general representations from large datasets, while fine-tuning adapts these representations to a specific task, leading to faster convergence, better generalization, and improved performance.

(d) LSTM is an excellent model for solving sequence-to-sequence problems. It contains multiple components such as forget gate, input gate, output gate, etc. Briefly describe the functionality of forget gate and input gate.

[6 marks]

(e) Most the state-of-the-art deep architectures consist of pre-training phase followed by fine-tuning phase. Please briefly describe what is pre-training and fine-tuning and Why it helps with the model's performance.

[6 marks]